



REPENSER A FAIRE LE LIEN AVEC ANDREAS PAR RAPPORT A SON DERNIER COURS "SC & VSM" qui évoluera probablement ; + vers récap des outils du Lean

# - INDUSTRIAL ORGANIZATION - Lean Management

You're a violinist and you're seeking for perfectly perform a very difficult concerto.

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# Introduction to the Lean thinking



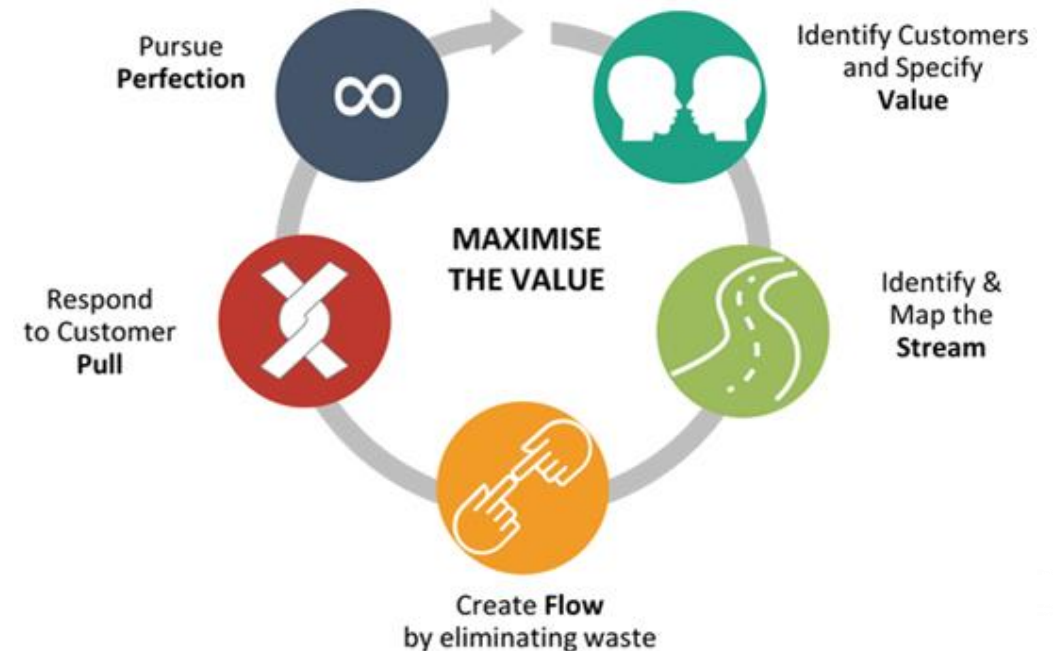
W. Edwards Deming  
"Continuous improvement"



1930s

1950s

Diagram by Kam G. Bultuk (<http://www.bultuk.com>)





# Course goals

- Understand the history of lean
  - Know the principles of Lean Management and Methodology
    - Provide an overview of the main tools used in Lean management



# Agenda

1. Presentation of **Lean Management**
2. Lean management **principles**
3. Lean management **methodology**
4. Lean **tools**

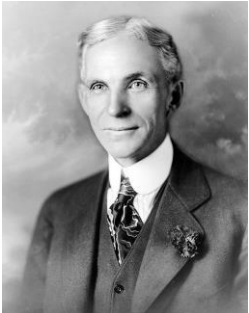


# **1. Presentation of Lean**



# 1. Presentation of Lean

## History of Lean



Henry Ford

### 1890 - Frederick Taylor > Taylorism

- scientific organisation of work (division and simplification of tasks), optimal manufacturing method (one best way), merit pay (fair's day work), primary productivity goal (mass production)

### 1913 - Henry Ford > Fordism

- standardization in mass production, assembly line work, reduction of wastes, improved Supply Chain Mngt

### 1962 - Toyota Production System (TPS) > Toyotism

- Sakichi Toyoda, Kiichiro Toyoda, Taiichi Ōno\*, Eiji Toyoda: the Toyota Production System (TPS) was born > Kanban, Total Quality Management, Reduction of wastes (muda, muri and mura), SMED, Poka-Yoke, Just-in-Time method, Kaizen, Genba, Andon, Jidoka, Heijunka...



Taiichi  
Ōno

### 1987 - Progressive adoption in the West > Lean

- John Krafcik: global management method > birth of the term « Lean »

### Since 2001 - Further developments

- The Toyota Way: continuous improvement and respect for people > 14 principles of Lean management



# 1. Presentation of Lean

## Other concepts and new fields of application

Lean, originally designed for production ("**Lean Manufacturing**"), has developed in several other environments, including:

- "**Lean office**" or "**Lean service**": deploys the principles of Lean in office activities or processing intangible flows (rather than physical products)
- "**Lean Engineering**" or "**Lean Development**" in Research and Development or Engineering processes
- "**Lean Software Development**" or "**Lean IT**" in IT development
- "**Lean Start up**": create a company more quickly and to deliver the right product to customers earlier and on the first try.
- Other concepts have been added, such as six-sigma, to form "**Lean Six-sigma**": an approach based both on the voice of the customer (surveys...) and on measurable and reliable data (by indicators)



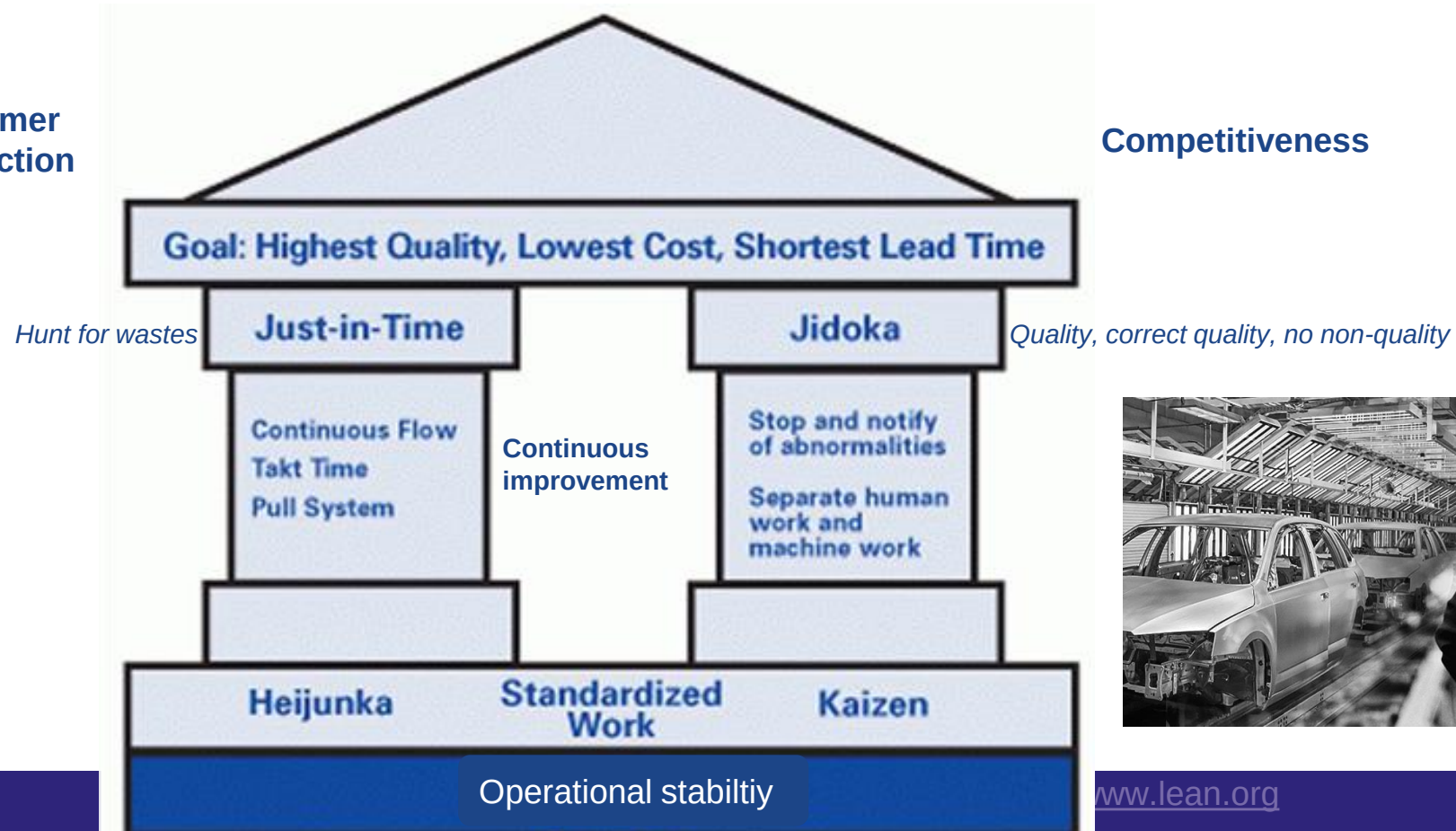


# 1. Presentation of Lean

## The Toyota Production System - TPS and the Lean temple

Customer  
satisfaction

Competitiveness





## 2.The **Lean management** principles



## 2. Definition of Lean

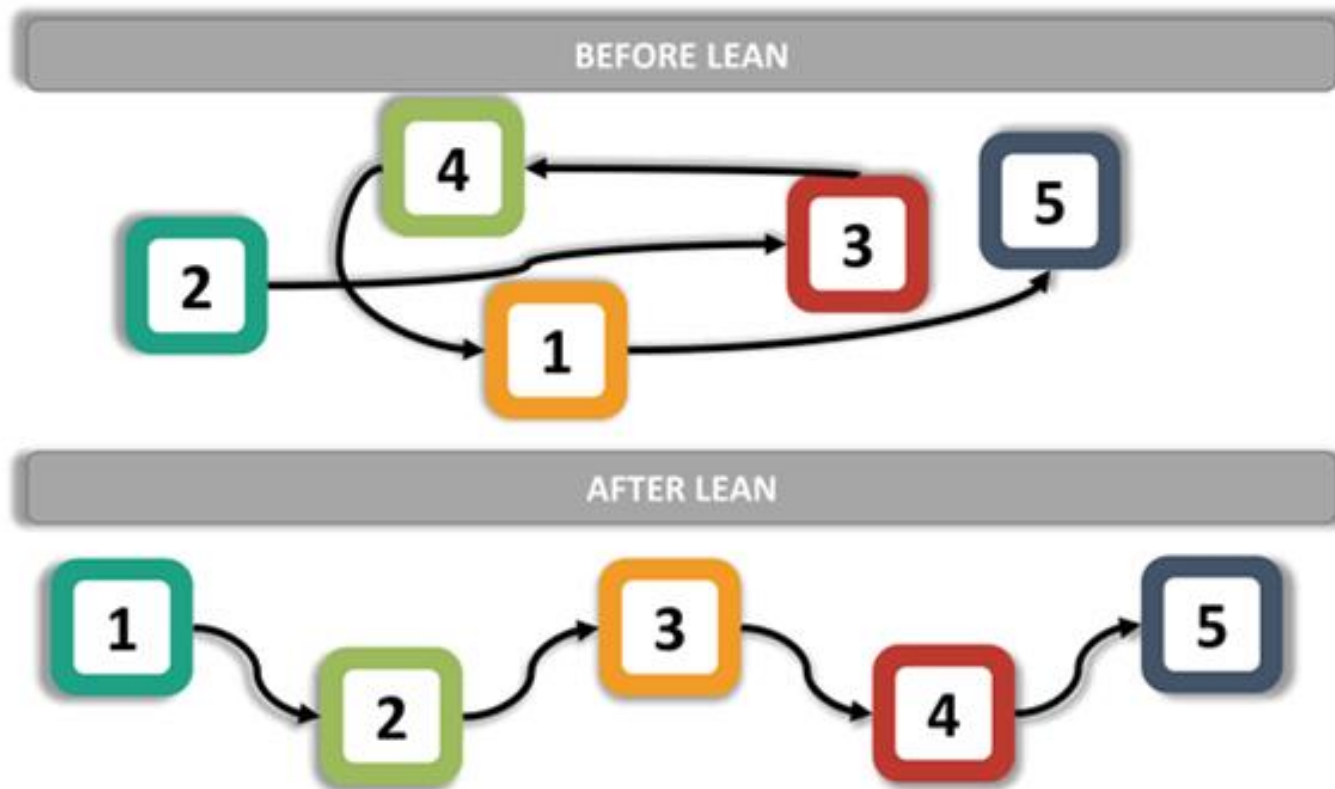
APICS Synonyms: Lean Management / Lean Manufacturing / Lean / Lean Production:

- A philosophy of production that emphasizes the minimization of the amount of all the resources (including *time*) used in the various activities of the enterprise.
  - It involves identifying and eliminating non-value-adding activities in design, production, supply chain management, and dealing with customers.
- Lean producers employ teams of multiskilled workers at all levels of the organization and use highly flexible, increasingly automated machines to produce volumes of products in potentially enormous variety.
- It contains a set of principles and practices to reduce costs through the relentless removal of wastes and through the simplification of all manufacturing **and** support processes.

(APICS Dictionary)

## 2.The **Lean management** principles

Before and after Lean Management: a typical output of a Lean implementation



Reduction of wastes

Simplification

Standardization

## 2. The **Lean management** principles



### IDENTIFY VALUE

Answer to critical questions: What is the **customer willing to pay for**? What is the value from the **customer's perspective**? What is the value the company wants to deliver?



### MAP THE VALUE STREAM

Map the workflow of your company (**Value Stream Mapping**), including all actions & people involved in **delivering the end product** to the customer. Understand and analyse the workflow. Identify the **wastes** within the steps.



### CREATE CONTINUOUS FLOWS

Ensure that each team's workflow remains **smooth**. This may take time, and it involves cross functional teamwork. **Bottlenecks** & interruptions will appear: detect them and remove them as much as possible, **step by step**



### CREATE A PULL SYSTEM

Introduce a pull system between the steps for which a **pull system is relevant**. Therefore, production is being launched if and only if there is a customer **demand**. Think of a **pizza**!



### SEAK FOR PERFECTION

Through **continuous improvement**, and Lean tools, always measure, analyze, and take actions in order to always **decrease the lead time and the amount of information** needed to serve the customer.



## 2.The **Lean management** principles

- The methodology and principles of Lean Management have a high level of focus on the **elimination of all the non-added value** activities
- Lean Management is really to be seen as a **management technique** that is most of the time focused on reducing the **wastes** generated within an organization, that negatively impact the **efficiency and performance** of the organization
- Definition of waste:
  - Anything that doesn't contribute to reaching the goal, anything that **doesn't add value** to a process

# 2.The **Lean management** principles: The 8 wastes of Lean

To remember The 8 Wastes, you can use the acronym “DOWNTIME.”

|   |                     |                                                                                    |                                                                                     |                                                                                      |                                                                                      |
|---|---------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| D | Defects             |   |   |   |   |
| O | Overproduction      | <b>Defects</b>                                                                     | <b>Overproduction</b>                                                               | <b>Waiting</b>                                                                       | <b>Non-Utilized Talent</b>                                                           |
| W | Waiting             | Efforts caused by rework, scrap, and incorrect information                         | Production that is more than needed or before it is needed.                         | Wasted time waiting for the next step in a process.                                  | Underutilizing people's talents, skills, & knowledge.                                |
| N | Non-Utilized Talent |                                                                                    |                                                                                     |                                                                                      |                                                                                      |
| T | Transportation      |  |  |  |  |
| I | Inventory (Excess*) | <b>Transportation</b>                                                              | <b>Inventory (Excess*)</b>                                                          | <b>Motion</b>                                                                        | <b>Extra-Processing</b>                                                              |
| M | Motion              | Unnecessary movements of products & Materials.                                     | Excess products and materials not being processed.                                  | Unnecessary movements by people (e.g. walking).                                      | More work or higher quality than is required by the customer.                        |
| E | Extra-Processing    |                                                                                    |                                                                                     |                                                                                      |                                                                                      |

*\*Be careful on one thing: inventory can be seen as an asset, as well as it can be seen as a waste. Inventory IS an asset, as long as it is justified. It's better to talk about smart inventory than just see the inventory as a waste*





## 2.The **Lean management** principle

### Reduce:

- stocks
- production cycles
- lead times

### Increase:

- productivity
- quality
- customer service

### ⇒ In order to reduce losses:

- of time
- of material
- of money

Lean: a **simple and effective approach** that improves productivity and reinforces the culture of continuous progress



## 2.The **Lean management** principle

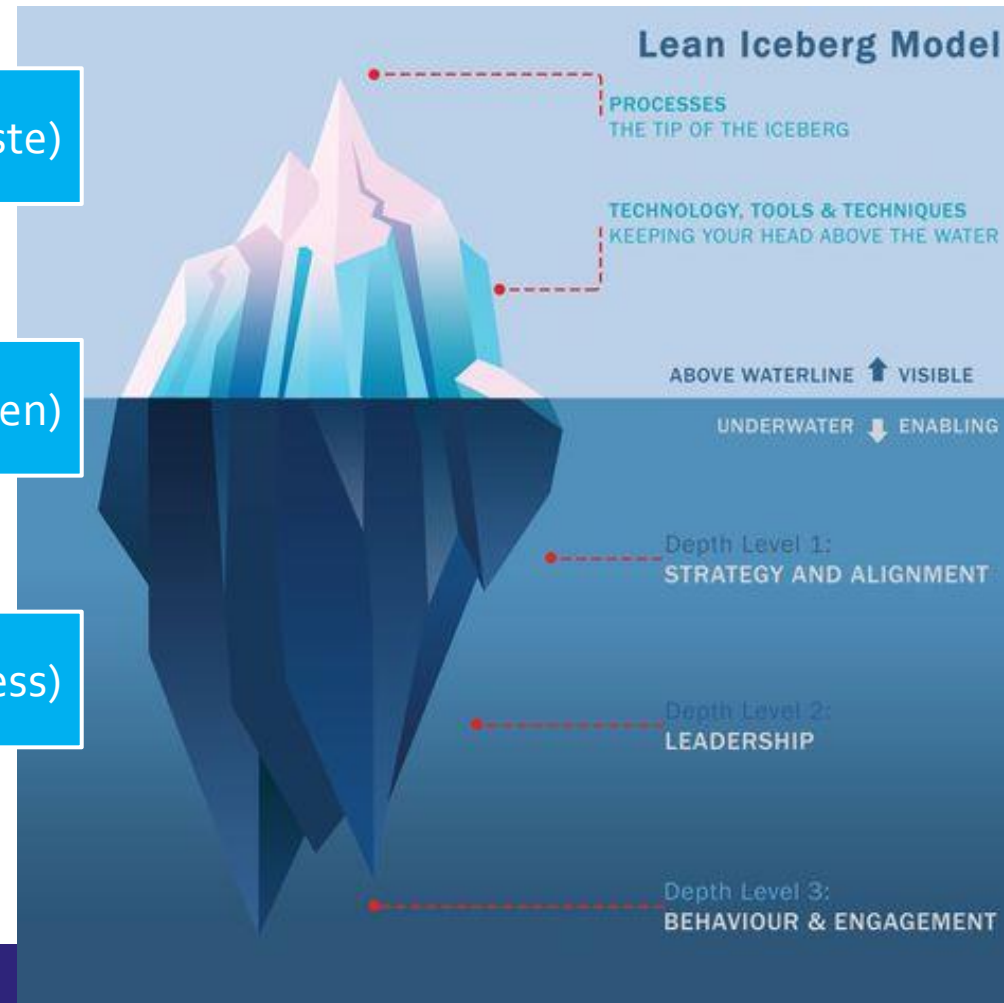
Eliminate the three types of deviations with the Lean management: **The 3 M's**



Muda (無駄, waste)

Muri (無理, overburden)

Mura (斑, unevenness)

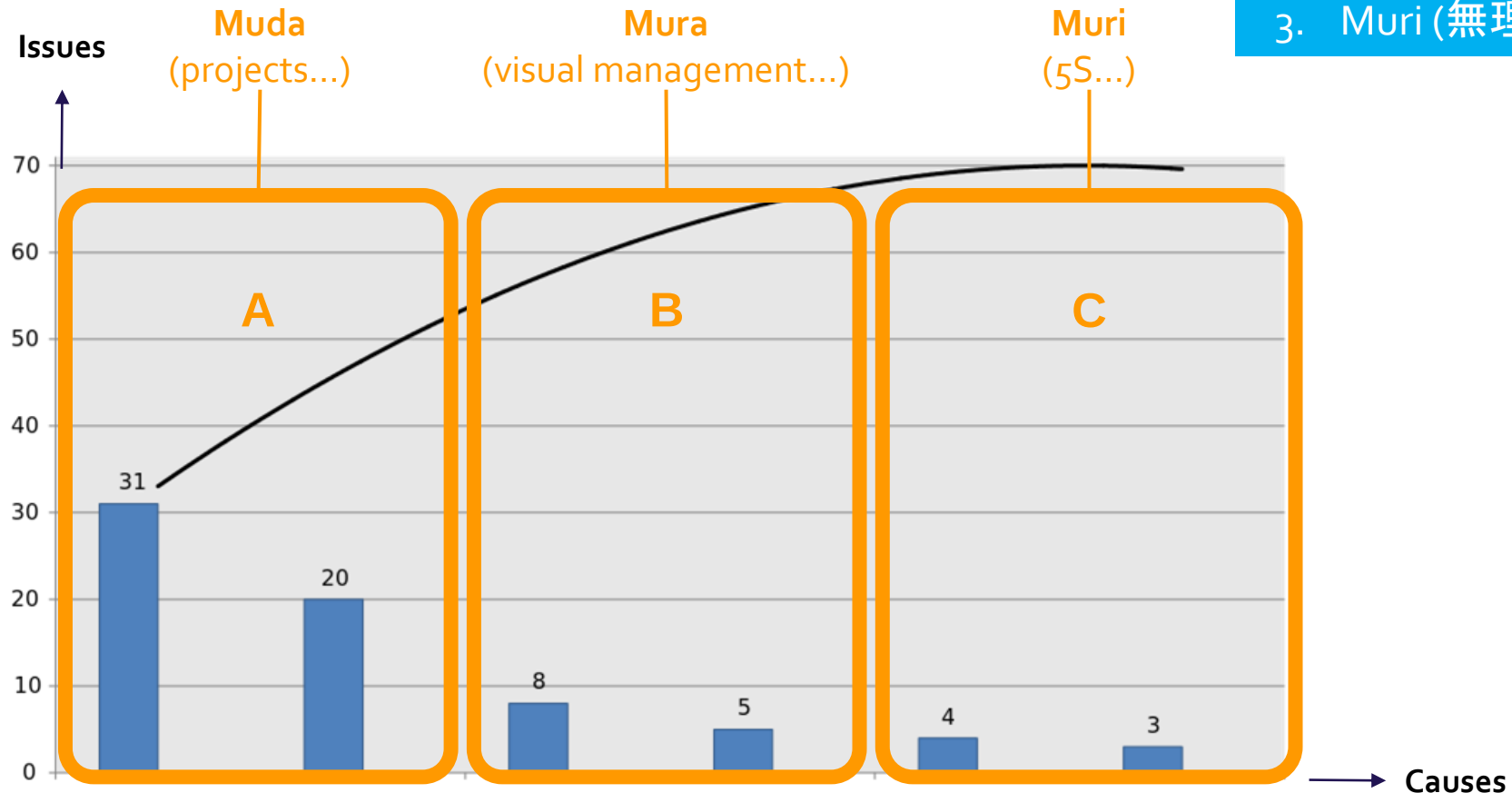


## 2.The **Lean management** principle

Eliminate the three types of deviations with the Lean management: **example with Pareto chart (80/20)**

The 3 M's:

1. Muda (無駄, waste)
2. Mura (斑, unevenness)
3. Muri (無理, overburden)



# 2.The **Lean management** principles

## A- Muda (無駄, waste)

Deliberately **accepting a margin of error, a failure rate, a level of non-quality** is in other words "generating MUDAS". By Muda, we mean **any deliberate waste** that we see and refuse to eradicate. These are the losses experienced on a daily basis and that we must relentlessly track down and eliminate.

### Characteristics of MUDA:

- "Loss" means anything that is unusable, unrecoverable or of no value;
- In manufacturing: "**Scrap**", products that have defects and are deliberately accepted;
- An activity that "consumes resources" and is ultimately "useless and superfluous", but is nevertheless maintained;
- Lost **investment**

### Examples of MUDA:

- Handling and transport activities (no need for conveying) ;
- Inadequate processing in manufacturing (Uses of defective, unnecessary parts) ;
- Unnecessary movements that can be dispensed with;
- Manufacturing defects (non-quality).



## 2.The **Lean management** principles

### B- Mura (斑, unevenness)

The aim of this approach is to maintain a regular, smooth and standard process flow at all times, free of waiting lines, bottlenecks and waiting times at workstations on the production line.

#### **Characteristics of MURA:**

- Form of waste "suffered";
- Whose source, reason or cause is elsewhere;
- Sometimes characterized by repeated interruptions in a workflow;
- To remedy it, the real cause must be unmasked, addressing the true source of the problem.

#### **Example of MURA:**

You notice an unusual increase in oil consumption by an engine, yet the number of hours of daily operation has not changed. If you decide to increase the stock of oil to prevent the risk of breakdowns, you create Muras. Instead, look for leaks and repair them.



## 2.The **Lean management** principles

### C- Muri (無理, overburden)

Muris constitute a waste of cash (fixed assets, overcharges and financial losses); a waste of space (occupation of storage areas by material that may not be used, risk of obsolescence if the excess material is not consumed in time). The abundance of resources always ends up becoming harmful.



### **Characteristics of MURI**

- Unsubstantiated" form of waste;
- Caused by adherence to inappropriate policies or standards;

### **Examples of MURI**

- Equipment and materials purchased in excess, in advance, as a result of miscalculated forecasts, a purchasing policy or unsuitable lot sizes,
- Equipment purchased but almost never used;
- Excessive labor, poorly employed, underemployed;
- Unnecessary operations resulting from inappropriate operating procedures.



# Quick break

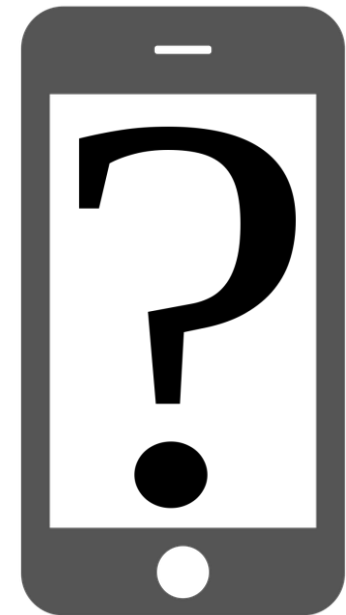
1. Go to the website with smartphone or computer:

[www.menti.com](http://www.menti.com)

2. Add your First and LAST name (e.g. DUPONT Martin)

3. Input the code **22 66 24 64**

4. Answer to the questions





# 3. Lean management methodology



# 3. Lean management methodology

## 1. Lean approach

### Lean thinking:

- “With **less** you can do a lot more!”
  - “With **all** we can do much better!”

Lean management, beyond being a set of methods, is based on the **human factor**. It suggests that the staff works in a state of mind oriented towards the reduction of waste and losses (of time, materials, money ...).

**Motivation and human behaviour** are necessary for effective application.



# 3. Lean management methodology

Nothing can be done without people



<https://www.sketchbubble.com>



# 3. Lean management methodology

## 1. Lean approach

The LEAN approach aims at stabilize a sustainable system by implementing 3 elements :

1. Strategy
2. Production system
3. Change of behaviour



# 3. Lean management methodology

## 1. Lean approach

### 1. Strategy:

- An **operational** strategy that focuses on **speed of action** and **reduction of delays** to create a sustainable competitive advantage
- Usually, this comes from a **vision** of the top management about the company's strategy, roadmap and priorities



# 3. Lean management methodology

## 1. Lean approach

### 1. Production system:

- An **efficient** production system to improve quality and reduce costs by **eliminating waste**
- Which delivers the products and services on **time**, in **full**, at the right **place**, according to the **customers' requirements**



# 3. Lean management methodology

## 1. Lean approach

### 1. Change of behaviour:

- An implementation philosophy and methodology based on a team spirit.
- A proactive approach to continuous improvement
- A will from management to empower people and give more responsibilities on the field



# 3. Lean management methodology

## 1. Lean approach

### Reduced costs

- Direct cost
- Trade-in cost
- Labour costs
- Cost of overtime

### Lead time reduction

- Program delay
- Production time
- Meeting deadlines
- Stock level

### Improvement in quality

- Loss rate
- Correction time
- Good on the first try
- Empowered and motivated teams
- Customer satisfaction, company image

# 3. Lean management methodology

## 2. Conditions for success

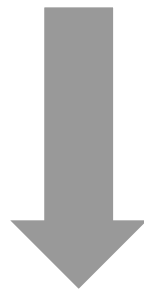
- A combined Top-down / Bottom-up actions and
- A perfect mastery of methodological tools

### Top-down

#### The role of top management:

- Motivate the project team
- Giving a sense of urgency and responsibility
- Communicate on expected objectives

Focused and concentrated efforts over short periods (3 months)

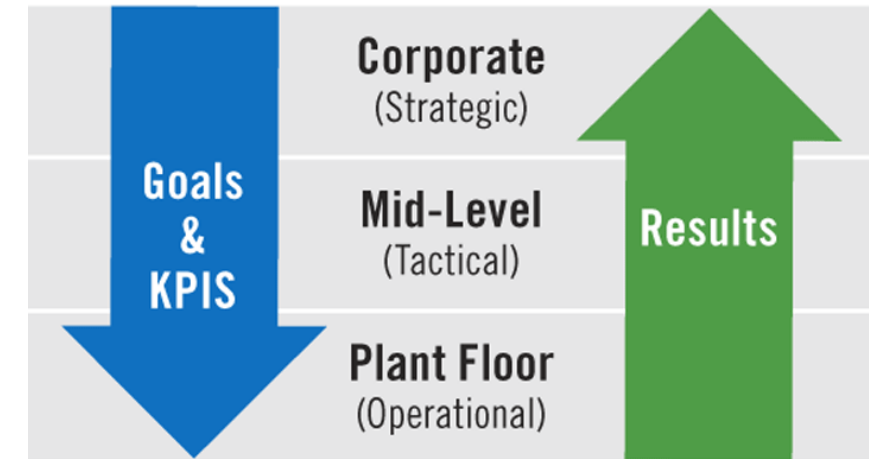


### Bottom-up

#### Optimal operation:

- Collaboration between functions
- Exchange of ideas and team spirit
- Methodical search for the causes of malfunctioning

Strong participation and mobilization of the field



<https://www.leanproduction.com>



# 4. Lean tools





# 4. Lean tools

## Several tools:

- 5S
- Andon
- Jidoka (Intelligent automation)
- Gemba
- Heijunka (Level Planning)
- Just-In-Time (JIT)
- Kanban (Pull System)
- Kaizen (Continuous Improvement)
- PDCA
- KPIs (Key Performance Indicators)
- Takt Time
- Total Productive Maintenance (TPM)
- Value Stream Mapping (VSM)
- SMED
- Poka-Yoke
- ...



# 4. Lean tools

5S



The 5S method is based on the principle that efficient, high-quality work must necessarily be carried out under optimal conditions of cleanliness, safety and rigour.

The 5S are Seiri, Seiton, Seiso, Seiketsu, Shitsuke and refer to 5 actions:

- Seiri: Sort
- Seiton: Set in order
- Seiso: Shine
- Seiketsu: Standardize
- Shitsuke: Sustain

The 5S method makes it possible to build a convenient working environment, the first three steps concern the conditioning of the land, the other two allow maintenance and improvement.







# 4. Lean tools

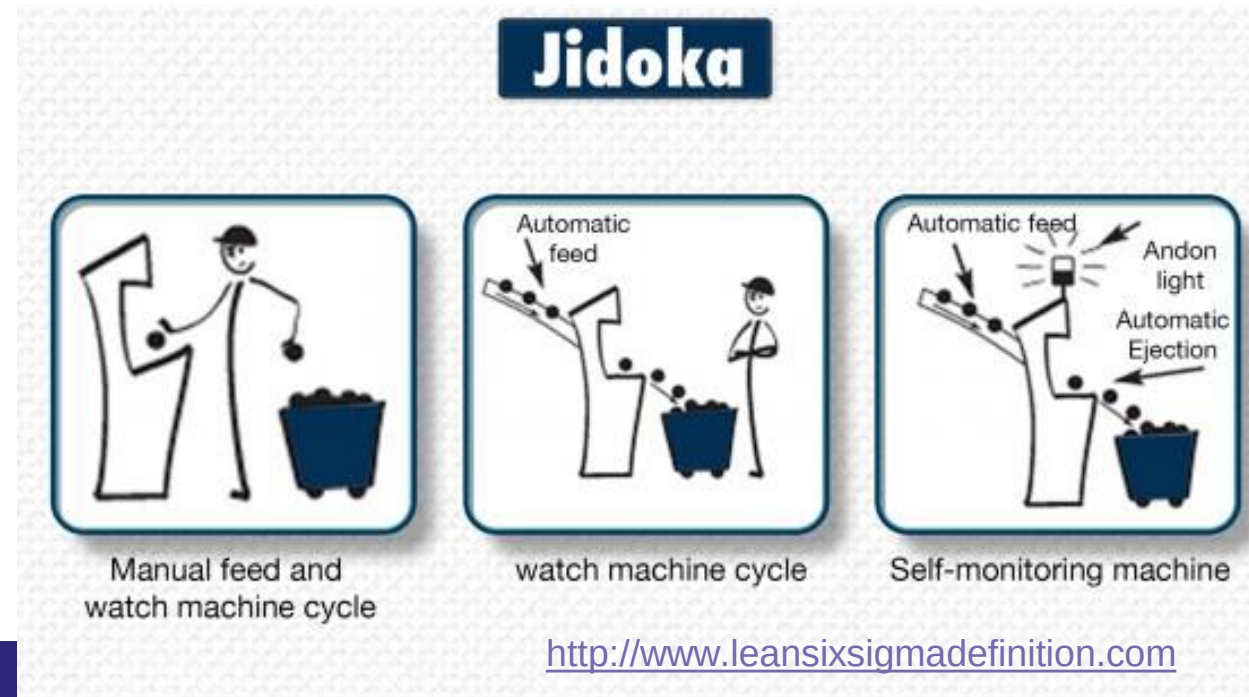
## Andon

A visual system (usually a light panel, cord, flag) to reveal production anomalies so that they can be addressed instantly.



## Jidoka (Intelligent automation)

Design equipment to partially automate the manufacturing process (partial automation is usually much less expensive than full automation) and to stop automatically when defects are detected. After Jidoka, workers can monitor several stations simultaneously (reducing labor costs) and many quality problems can be detected immediately (improving quality).



# 4. Lean tools

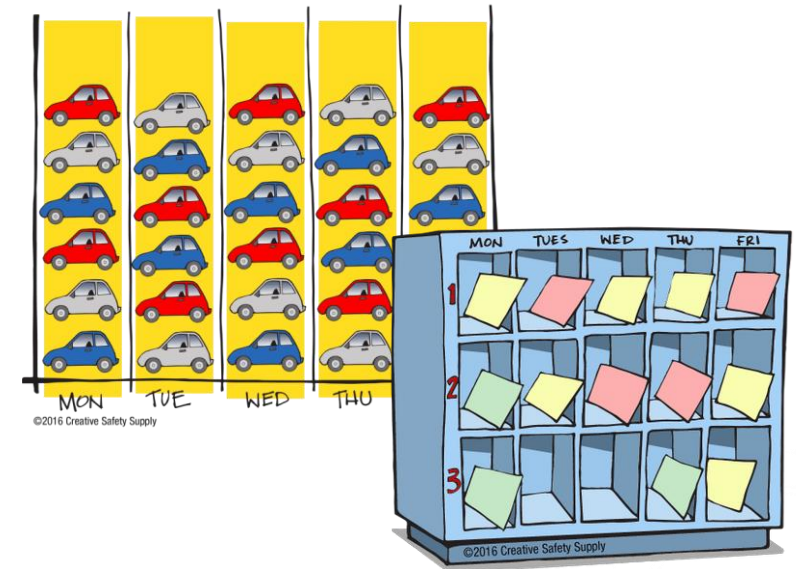
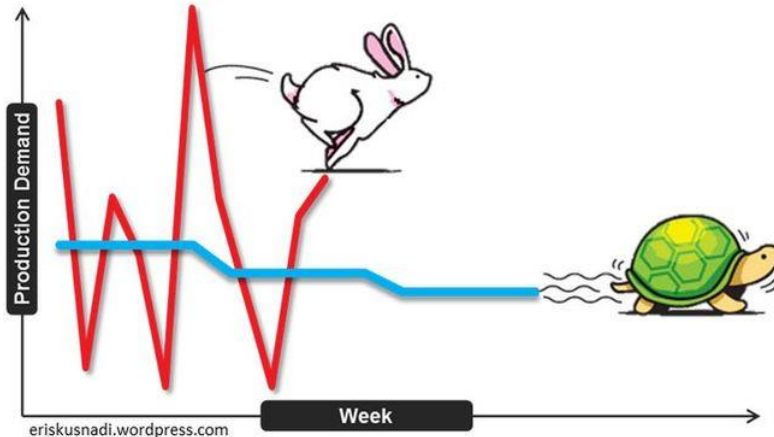
## Genba (or Gemba)

A philosophy that reminds us to get out of the office and spend time in the field - the place where real action happens. This fosters an in-depth understanding of real-world manufacturing issues through direct observation and interaction with employees.



## Heijunka (Level planning)

A form of production planning that aims at manufacturing in small batches by mixing several product variants in the same process. It reduces lead times (since each product or variant is produced more frequently) and inventory (since the batches are smaller).







# 4. Lean tools

## Just-In-Time (JIT)

Working "Just-In-Time" means working without unnecessary inventory. Streamline the production process based on customer demand instead of making it more cumbersome based on projected demand. This is supported by numerous lean tools, such as Continuous Flow, Heijunka, Kanban, Standardized Work and Takt Time. Just-In-Time is very effective in reducing inventory levels, improving cash flow and reducing space requirements.



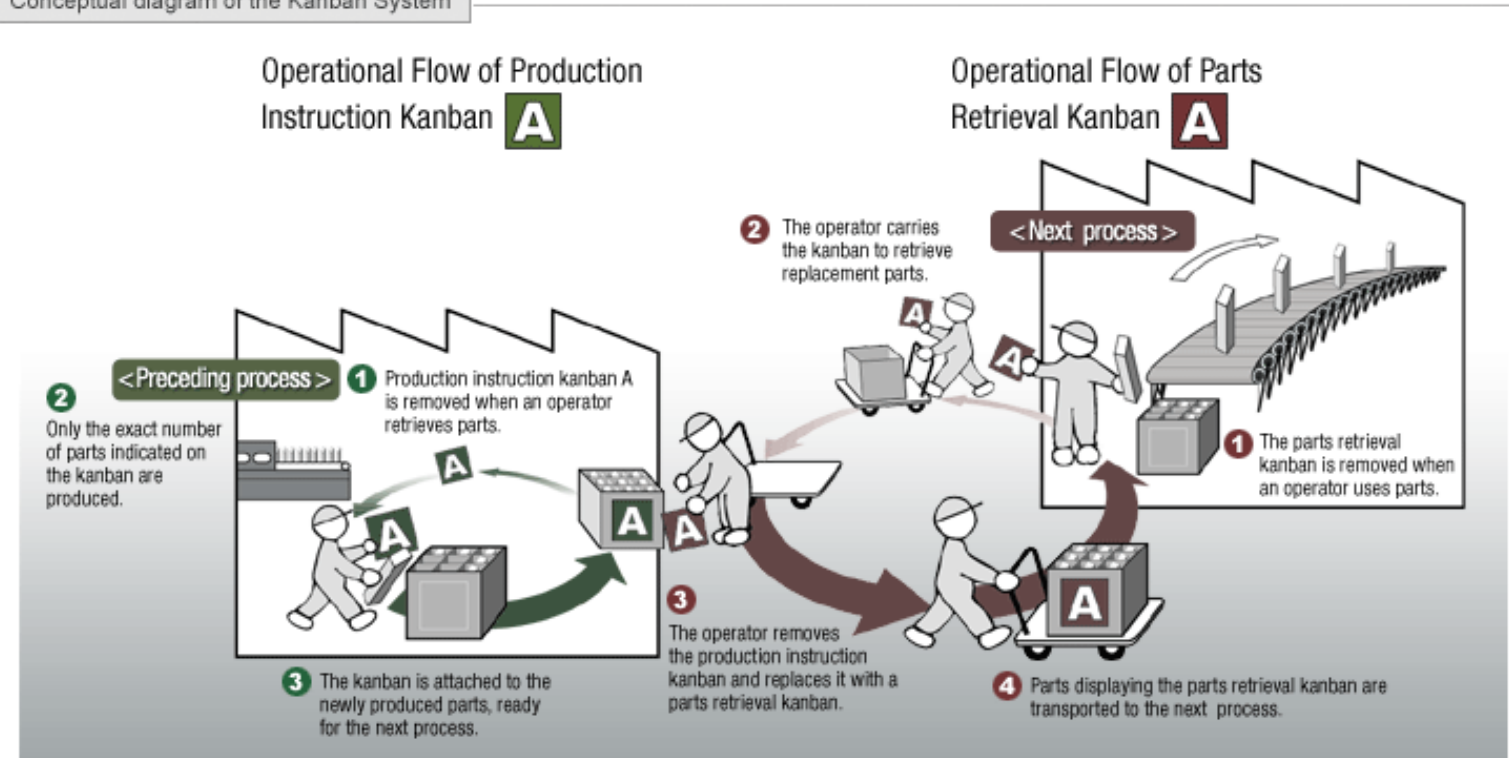
© pixabay / Femme Actuelle

# 4. Lean tools

## Kanban (Pull System)

A method of regulating the flow of goods both within the plant and with external suppliers and customers. This method imposes an automatic replenishment request through notifications sent to the storage facility. It eliminates inventory waste and overproduction. Pull and push flows can coexist in the same company. Nowadays, Kanban is more and more also handled in information Systems

Conceptual diagram of the Kanban System



[v.leansixsigmadefinition.com](http://v.leansixsigmadefinition.com)

# 4. Lean tools

## Kaizen (Continuous Improvement)

Align corporate objectives (Strategy) with middle management plans (Organization) and field work (Operation). This ensures that actions in the field are consistent with the strategic objectives and eliminates wastage due to miscommunication between management and operations.

## PDCA

An iterative methodology for the implementation of lean manufacturing :

- Plan (establish the plan and the expected results)
- Do (execute execution plan)
- Check (check the results obtained)
- Act (review and evaluate, do it again)

### Benefits of Kaizen

Increased productivity



Improved quality



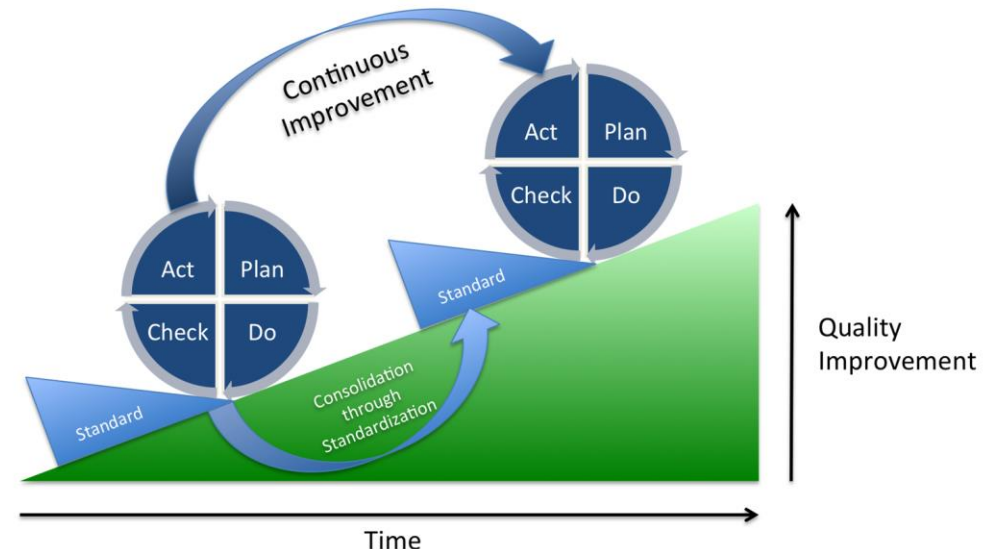
Higher morale & employee satisfaction

Better safety



Improved communication

<https://tulip.co/resources/>





# 4. Lean tools

## KPIs (Key Performance Indicators)

Measures designed to monitor and encourage progress towards the organization's critical objectives.

The best production KPIs:

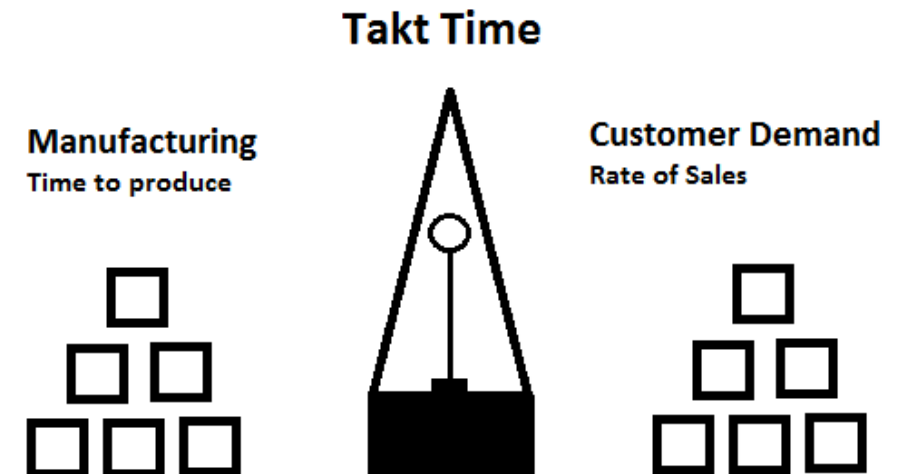
- Are aligned with high level strategic objectives (thus contributing to the achievement of these objectives)
- Are effective in exposing and quantifying waste (OEE is a good example)
- Are directly related to the performance of the plant's employees (so that they can generate results)

## Takt Time

A lean method that aligns the production rate with customer demand. It is defined as the key figure:

Available Production Time / Customer Demand.

Takt Time provides a simple, consistent and intuitive method of production scheduling. It can be easily extended to provide an efficiency target for the plant (Actual Parts / Target Parts).

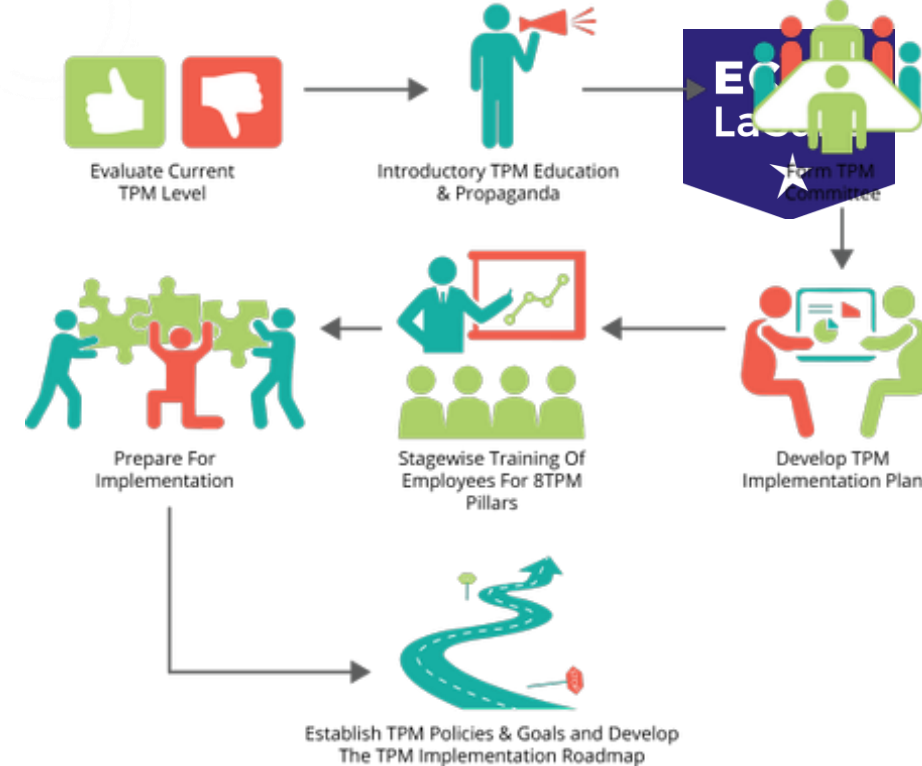




# 4. Lean tools

## Total Productive Maintenance (TPM)

A proactive and preventive maintenance improvement approach to maximize equipment uptime. TPM blurs the separation between maintenance and production by encouraging operators to continuously maintain their equipment. This approach creates a shared responsibility for the equipment among employees and encourages their involvement.

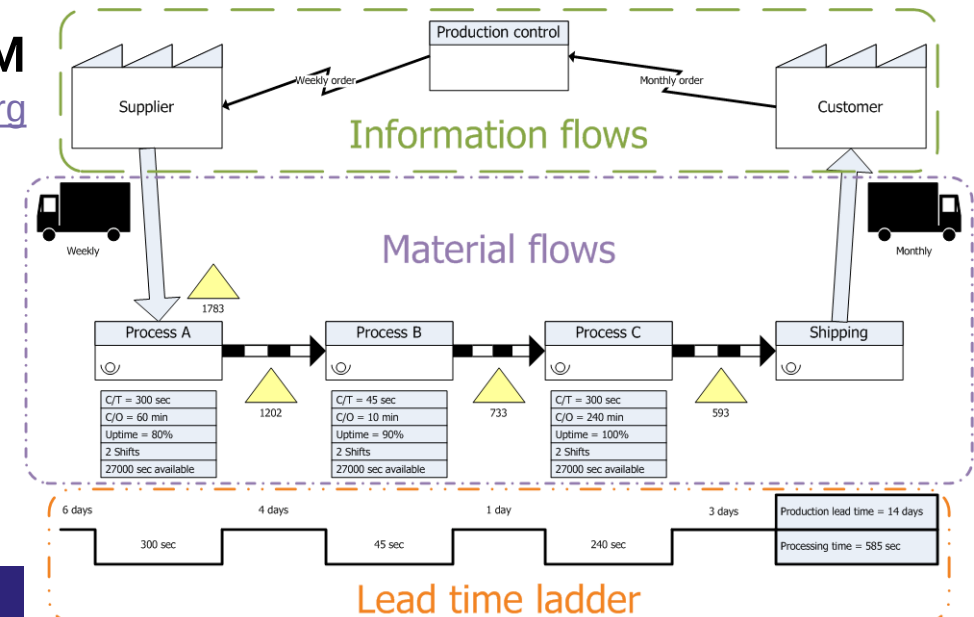


## Overview of VSM

<https://en.wikipedia.org>

## Value Stream Mapping (VSM)

A tool used to map production flows visually. Indicates the current and future state of processes in a way that shows opportunities for improvement. VSM exposes waste in current processes and provides a roadmap to converge towards an improved future state.





# 4. Lean tools

## The SMED method (Single Minute Exchange of Die)

The SMED method consists of reducing the setting times to a minimum. The steps are as follows:

1. Making the adjustment steps as external as possible, i.e. they can be carried out during production (machines in operation).
2. Simplify internal adjustments (e.g. replace screws/bolts with handles and levers)
3. Eliminate non-essential operations
4. Create standard work instructions

It promotes small batch manufacturing, which reduces inventory and improves customer agility.

## Poka-Yoke

Detection and prevention of design errors in production processes with the aim of having zero defects. This comes from the realization that it is difficult (and costly) to find all defects by inspection and the correction of defects usually becomes much more expensive at each stage of production.



<https://www.tjo.ee>



# 4. Lean tools

## Other tools exists...

- SMART goals
- Visual Management
- Standardization
- Hoshin Kanri
- ...



# Bonus: application of **Lean** with Sheldon Cooper (*The Big Bang Theory*)





# Bibliography

## To go further:

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# /!\ Quick assessment /!\

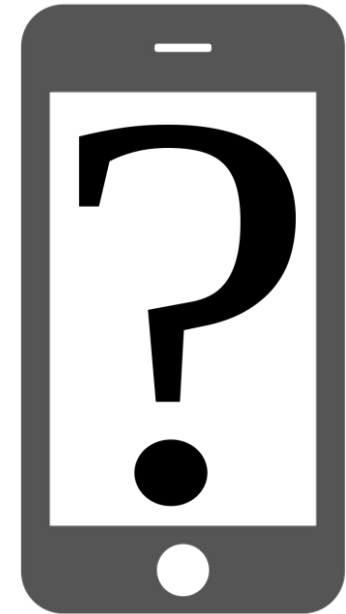
1. Go to the website with smartphone or computer:

[www.menti.com](http://www.menti.com)

2. Add your First and LAST name (e.g. DUPONT Martin)

3. Input the code **22 66 24 64**

4. Answer the questions



# Thank you for your attention



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